

CareOneX Voice: a voice AI agent that helps families find and pay for homecare

Team: Homecare agents

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1. Statement of Work

New Jersey has more than 1.5 million residents aged 65 and older, and most of them want to stay in their own homes as they age. Home care makes that possible, but paying for it is confusing. Medicaid's Managed Long Term Services and Supports (MLTSS), Medicare home health, the state's JACC program, veterans' benefits and private pay each have their own eligibility rules, application steps and vocabulary. The person navigating this is rarely the patient. It is usually an adult daughter or son trying to understand a system nobody explained to them at the moment a parent's needs have suddenly changed.

Many of these families start online. On **careonex.com**, a family fills out a short form describing the situation and asking for help. What has to happen next is a phone call: someone must call the family back, listen to what is really going on, ask the questions a form cannot, explain the options in plain words, and learn whether care is needed now, later, or not at all. Today that call depends on a person being free to make it: evening inquiries wait until the next day, weekend requests wait longer, and when volume rises some families are never called. Every unreturned call is a family left to work the system out alone.

This is the gap the project addresses. An AI voice agent can return the call within minutes of the form being submitted, at any hour, and hold a real conversation: ask about the parent's needs, answer questions about how care is paid for, and record what it learned for the intake staff who take over from there. Three developments make this feasible now. Streaming speech recognition, low-latency language models and neural text-to-speech can together sustain a phone conversation with sub-second turns. Retrieval-augmented generation (RAG) lets a model answer from a maintained corpus of program documents rather than from memory, which matters because eligibility rules change and a wrong answer to a family has a real cost. And tool-using agents in the ReAct tradition show how a model can carry out a structured task, here an intake, while still conversing naturally.

The same literature is clear about the hard part. Benchmarks such as *T-bench* show that agents which do well turn by turn still fail on multi-step tasks, and that reliability must be measured against simulated users rather than single prompts. For a voice agent talking to a stressed caregiver, reliability also means never inventing a fact and never giving medical advice, so the evaluation harness is built alongside the agent, not after it. The team is well placed to build it: CareOneX runs the intake page in question, so we know the questions families actually ask, the fields an intake must capture, and the consent language on the form that makes an outbound call permissible.

Problem statement

Can an outbound voice agent call a family that has filled out an online form about home care, hold a natural conversation that establishes whether and what kind of care they need, answer their questions about paying for care from cited sources, and hand back an accurate structured intake record?

2. Data sources

Three kinds of data: a knowledge base the RAG component retrieves from when a family asks a question; conversation data to design, prompt and evaluate the dialog; and speech data to test recognition on the voices we will actually hear. All sources are public or team-generated. The CareOneX intake form supplies its field vocabulary and consent language.

2a. Knowledge sources for the RAG (what a family asks on the phone)

Chosen from the family's point of view: how care is paid for, who qualifies, how to apply, what care includes, how to judge a provider. Every document carries source URL, program, publisher, effective date and fetch date, so an answer can name its source and a stale page can be found.

Source (HTML and PDF, a few hundred pages)	What it contains	Why it matters on the call
NJ FamilyCare / Medicaid MLTSS pages, member handbooks, DMAHS FAQs	Clinical and financial eligibility, income and asset limits, covered in-home services, how to apply	"Does Medicaid pay for someone to come to the house?" is the most common question and the least understood
NJ Division of Aging Services: JACC, Statewide Respite, Alzheimer's Adult Day, PACE, county ADRC directory	Non-Medicaid state programs, who qualifies, what they pay for, the county office to call	Options for families above Medicaid limits, and a real number to give them
Medicare.gov home health pages; CMS booklet Medicare & Home Health Care; VA Homemaker/Home Health Aide and Aid & Attendance pages	What Medicare covers at home (homebound and skilled-need rules); veterans' benefits and how to apply	Corrects the belief that Medicare pays for long-term help; surfaces a path most eligible veterans never hear about

NJ Consumer Affairs Health Care Service Firm rules and consumer guide; NJ DOH licensed agency list; CMS Care Compare methodology	Who may legally provide home care in NJ, what a star rating means, what to ask a provider	Answers “how do I know an agency is legitimate?” and explains care types in plain words
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Re-crawled on a schedule; conflicting documents are resolved by effective date; pages that do not parse cleanly are excluded, because a half-read rule spoken aloud is worse than “I will have someone follow up on that”.

2b. Conversation and speech data

Dataset	Origin, format, key attributes	Role, and quality concerns with mitigation
Intake schema and consent language of the CareOneX family form	Public web form; coded fields (county, kind of help, hours per week, timeline, relationship, payer); the “you agree to be contacted” text	Defines what a successful call must capture and who may be called. No submitted records are used.
Family personas and call scripts (team-built)	300+ JSON personas: back-story, speaking style (hesitant, talkative, hard of hearing, non-native English), gold needs profile, disposition, questions they will ask	Drives the simulated-user harness. Risk of encoding our own assumptions: two members write, a third reviews.
Team-recorded calls	WAV at 8 kHz telephone quality through Twilio, hand transcripts, interruption timings	End-to-end test of recognition, barge-in and latency. Small by design.
Mozilla Common Voice (English), LibriSpeech	Public; about 3,000 and 1,000 hours; speaker age bands (60–79) and accent labels; validated transcripts	Recognition evaluation on older and accented voices, down-sampled and band-limited. Clean read speech, so it understates real difficulty.
MultiWOZ 2.2, Schema-Guided Dialogue; CMS Home Health Care Compare	Public task-oriented dialogue corpora (JSON); federal agency table (CSV, quarterly)	Few-shot examples of slot filling and repair (form, not content); a list of licensed agencies the agent can offer to text. Ratings lag a quarter; the agent says so.

3. Scope and preliminary design

Course components. We commit to three: **complex models showcasing MLOps** (a streaming pipeline of speech recognition, an LLM dialog agent with RAG and tools, and speech synthesis, with an evaluation harness, versioned prompts and data, CI/CD and monitoring); **computationally expensive inference** (three models per turn under a sub-second latency budget, so streaming, caching and model routing are the engineering core); **scalability** (many concurrent calls and the audio, transcripts and documents that come with them).

Objectives and success metrics

Objective	Metric	Target
The call sounds like a conversation	p95 turn latency (end of family speech to start of agent speech); barge-in	< 1.0 s; agent stops within 300 ms
Older and accented voices are understood	Word error rate on band-limited 60+ and accented test sets	< 15%
The intake record is right	Field and disposition accuracy vs. gold over the simulated-user set	≥ 90% each
Answers are grounded; no advice	Claims traceable to a retrieved chunk; medical or legal advice given	≥ 95%; 0
Nobody is called without consent	Calls to numbers without recorded consent; opt-outs honored	0; 100%
Production-grade delivery	Containers, tests and evals in CI, GCP deployment, monitoring dashboard	By the final milestone

Learning emphasis. The team knows Python and single-call LLM use and wants to learn real-time speech pipelines and telephony, agent orchestration with tools and retrieval, simulation-based evaluation of dialog agents, and the MLOps around it: Docker, data and prompt versioning, CI/CD, GCP (Cloud Run, Vertex AI, GCS/BigQuery) and monitoring of latency, cost and quality drift.

Agent pipeline and prompt engineering

A call moves through fixed stages. Each has one prompt, one input schema and one output schema, so it can be tested and improved on its own.

Stage	What happens	Prompt engineering and evaluation
0. Eligibility to call	A scheduler selects only records with recorded consent, inside permitted hours, not opted out, and dials through Twilio Programmable Voice (media streams).	No model on purpose: a rule the agent cannot override, covered by unit tests.
1. Opening	The agent gives its name, says it is an automated assistant calling for the service the family contacted, that the call may be recorded, and asks if now is a good time. “No” or “stop calling” ends the call and writes the opt-out.	Fixed script, not generated. Tested for exact wording and for every refusal phrasing in the persona set.

2. Listening	Streaming recognition with endpointing tuned for slower speakers; partial transcripts reach the agent as they arrive; barge-in cancels synthesis.	Endpointing thresholds and recognizer choice tuned on the recorded and band-limited sets; WER and latency per configuration.
3. Dialog agent	One LLM agent holds the state (a needs profile with a confidence per field) and picks the next move: ask, confirm, answer, hand off, close. One or two short sentences per turn.	System prompt with role, tone (warm, unhurried, eighth-grade language), state schema and move set; few-shot turns from persona scripts; read-back of names, counties and hours; hard rules: never diagnose, never promise eligibility, escalate on distress or confusion.
4. Answering (RAG)	On a question about paying or qualifying, the agent rewrites the query, retrieves (hybrid search filtered by program and date, re-ranked) and answers in two sentences naming the source, or promises a follow-up.	Chunking, embedding and re-ranker experiments scored on recall@k against gold chunks; a grounding prompt that forbids claims without a chunk; answers checked for length and reading level because they are spoken.
5. Speaking	Neural text-to-speech streamed sentence by sentence so the first words start before the answer is finished.	Voice, rate and pauses judged by a listening panel; time-to-first-audio tracked.
6. Wrap-up	The agent summarizes and confirms, offers to text licensed agencies nearby, closes. A post-call model writes the intake JSON, disposition and summary; a human reviews a sample.	Extraction prompt scored on field accuracy; disagreement between live state and post-call extraction is flagged.

Simulated users and prompt versioning. A second LLM plays each persona, speaks through text-to-speech and is heard through the same recognizer, so the agent is tested over the real audio path. Every prompt change replays the persona set in CI and is promoted only if field accuracy, disposition accuracy and grounding do not regress and cost per call stays within an agreed margin. Prompts are versioned files; the same harness runs our ablations: recognizer A vs. B, verifier on or off, frontier vs. open model.

Application mock design. The family's interface is the call itself, so the screens belong to the operator: a console with a call queue on the left (who is due, consent status, attempts, time window), the live call in the middle (streaming transcript, the agent's state and confidence per field, a "take over" button that hands the call to a person) and the outcome on the right (intake card, disposition, summary, sources cited, audio, review checkbox). A metrics screen shows reach rate, dispositions, latency, sampled WER and cost per call. Wireframes are attached to the final PDF.

Fun factor. It talks. Anyone on the team can call it, play a worried daughter in Monmouth County and try to break it; the persona harness turns evaluation into a game and the ablation leaderboard makes the engineering choices visible.

Limitations and risks

- **Consent and calling law come first.** The FCC ruled in February 2024 that AI-generated voices are "artificial" under the Telephone Consumer Protection Act, so the agent may only call people who gave prior express consent, must identify itself as automated, disclose recording, and honor a stop request at once. The project never cold-calls: it returns calls to families who submitted a request and agreed to be contacted. The demo uses team members' numbers and synthetic personas.
- A vulnerable population: the person answering may be confused, hard of hearing or in distress. The agent escalates to a human on any sign of it and never gives medical or legal advice or says someone qualifies.
- Recognition of older, accented, telephone-quality speech is the weakest link; we measure it on purpose and design the dialog to confirm rather than assume. Latency and cost (three models per turn) are managed with streaming, caching and routing; the fallback is a slower agent that says "let me check".
- Scope: New Jersey, home care, English, one call flow.

Milestones (tentative; course deadlines take precedence)

Milestone	Deliverable	Target
M1	Team registration; this Statement of Work	Sep 29
M2	Containerized ingestion of RAG corpus and speech sets; vector store; recognizer and synthesizer benchmarked; 100 personas; text-only agent completes an intake	mid Oct

M3	Live telephony end to end (recognition, dialog agent with RAG, synthesis, barge-in); 300-persona harness in CI; consent gate and opt-out; latency and cost baselines	early Nov
M4	Operator console; GCP deployment (Cloud Run), CI/CD, monitoring; load test with concurrent simulated calls	mid Nov
M5	Final demo with live calls, ablation results, report and video	early Dec

References. Yao et al. (2023) ReAct, ICLR · Lewis et al. (2020) Retrieval-Augmented Generation, NeurIPS · Yao et al. (2024) T-bench, arXiv:2406.12045 · Radford et al. (2023) Whisper, ICML · Zang et al. (2020) MultiWOZ 2.2; Rastogi et al. (2020) Schema-Guided Dialogue, AAAI · Ardila et al. (2020) Common Voice, LREC · FCC Declaratory Ruling, CG Docket 23-362 (Feb 8, 2024) · NJ DMAHS MLTSS pages · CMS Care Compare (data.cms.gov)